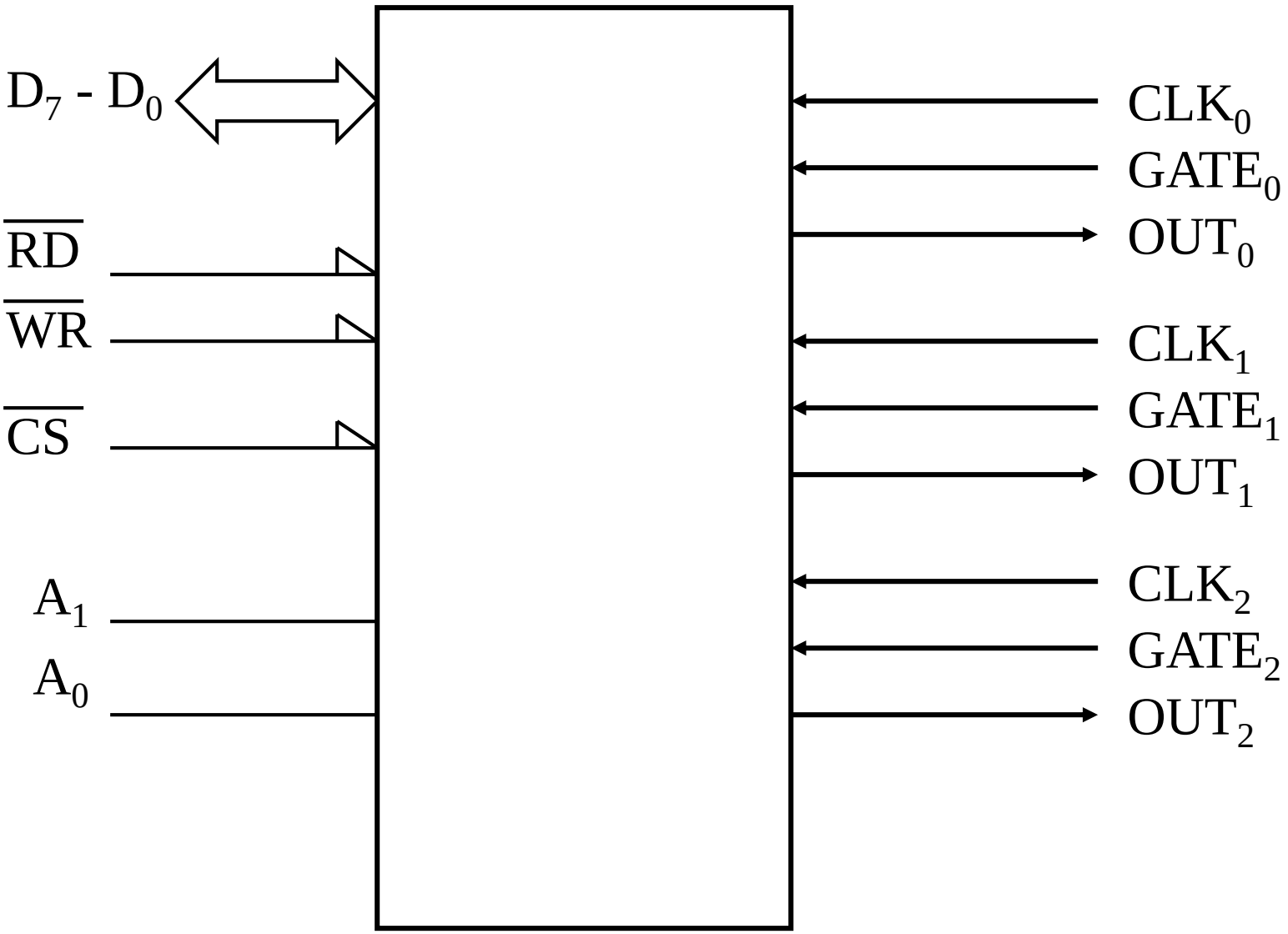
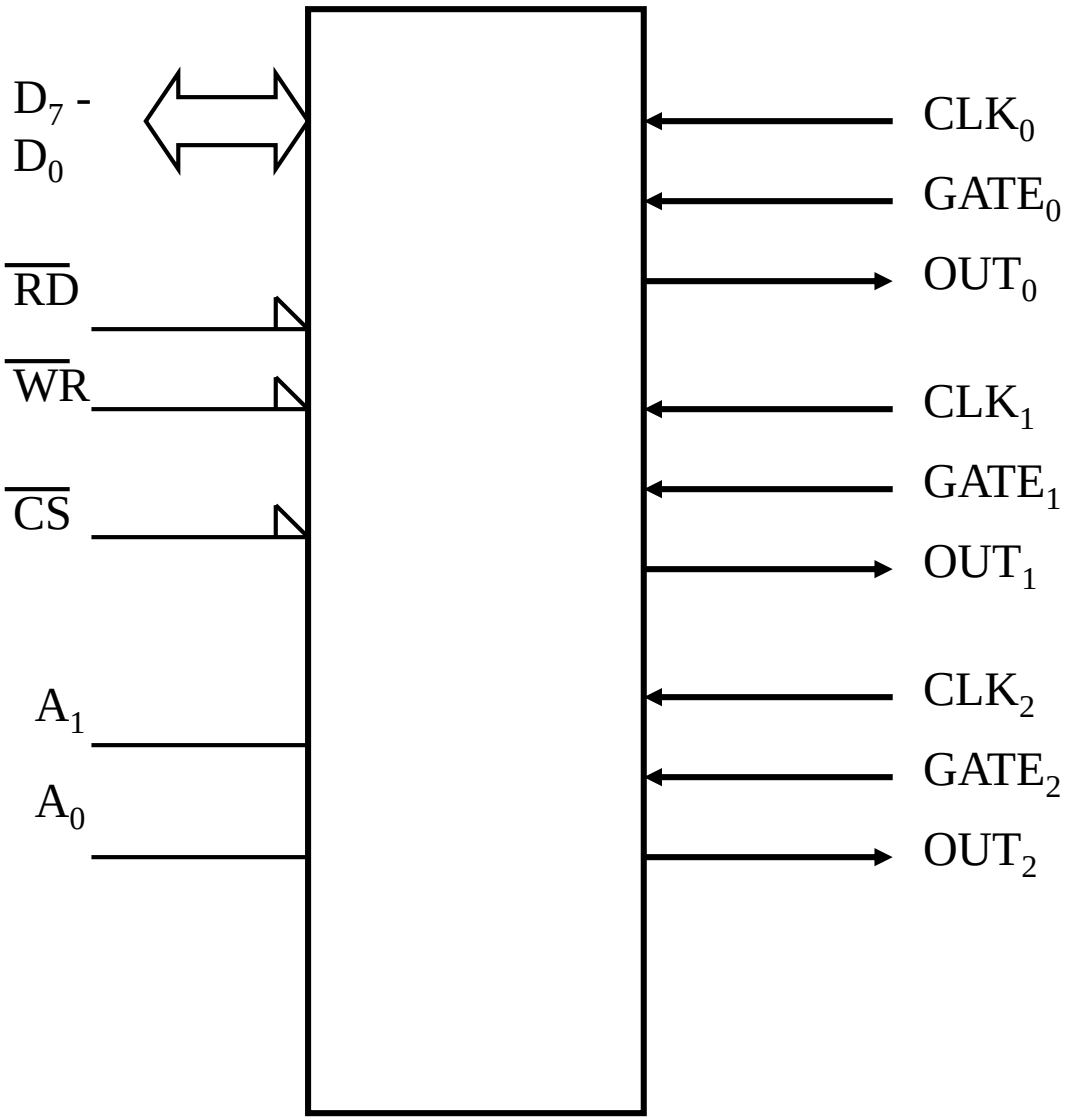
The 8253 Timer



The 8253 Timer



Data is communicated 8 bits at a time

2 address lines = 4 internal registers:

00 Counter 0

01 Counter 1

10 Counter 2

11 Control Word

8253 counters

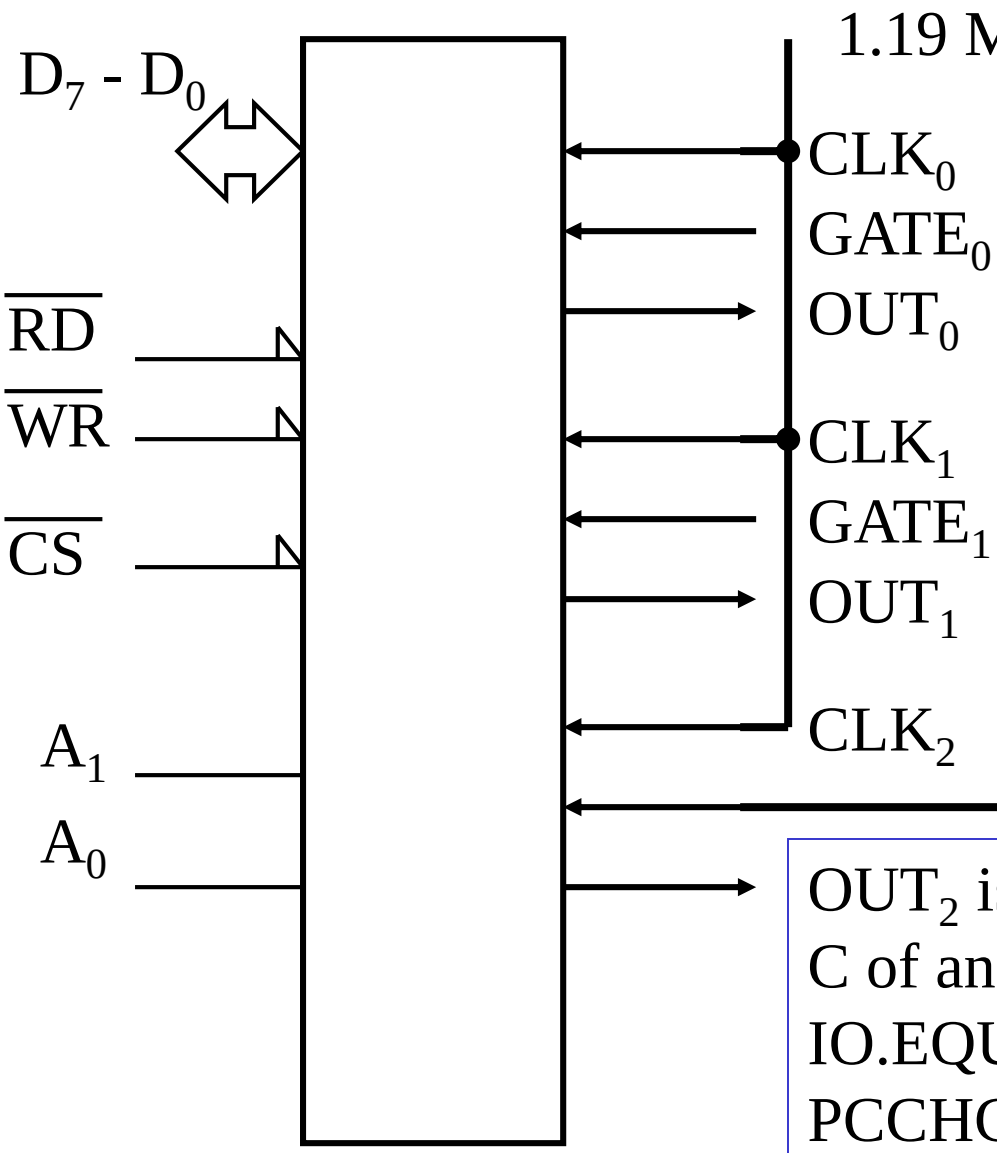
- Each counter is a 16-bit counter
- There are 8 data lines
- There is only one physical address per counter

- Therefore to write/read 16 bits into a counter requires a low-byte/high-byte write or read to the same address

8253 Control Word

8253 Control Word							
Select Counter		Read/Load		Mode			BCD
Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
=00 Counter 0 =01 Counter 1 =10 Counter 2 =11 Not allowed		=00 Latch Counter =01 Read/Load only most significant byte =10 Read/Load only least significant byte =11 Read/Load least significant byte, then most significant byte		=000 Mode 0 =001 Mode 1 =x10 Mode 2 =x11 Mode 3 =100 Mode 4 =101 Mode 5			=0 Binary Count 16 bits =1 BCD Count 9999 - 0

The 8253 Timer in the PC



$GATE_2$ is connected to bit 0 of Port B of an internal 8255. Using IO.EQU this can be accessed as PCCHB

OUT_2 is connected to bit 5 of Port C of an internal 8255. Using IO.EQU this can be accessed as PCCHC.

IO.equ

- IO.equ has equate statements that give readable names to the physical port addresses of peripherals for 485/486
- IO.equ can be included in every assembly by altering the Set a86 parameter

Set A86 = *mypath*\IO.equ+P64

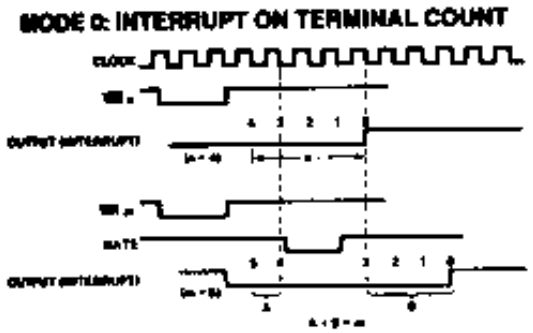
- Because IO.equ contains only equate statements which are evaluated at assembly time and do not generate executable statements, its presence will not affect the *.com program

Relation of gates to counters

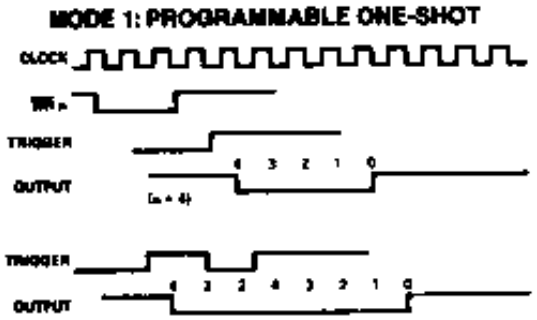
Signal Status Modes	Low Or Going Low	Rising	High
0	Disables counting	—	Enables counting
1	—	1) Initiates counting 2) Resets output after next clock	—
2	1) Disables counting 2) Sets output immediately high	1) Reloads counter 2) Initiates counting	Enables counting
3	1) Disables counting 2) Sets output immediately high	1) Reloads counter 2) Initiates counting	Enables counting
4	Disables counting	—	Enables counting
5	—	Initiates counting	—

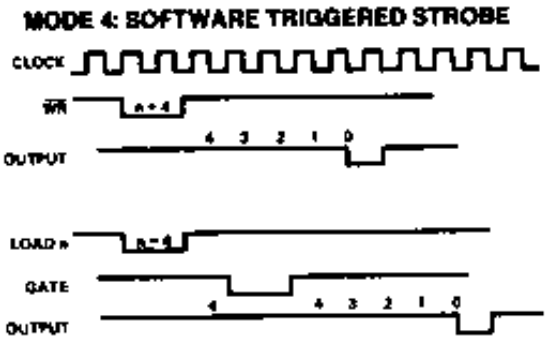
Taken from *The 8088 and 8086 Microprocessors, 2d Edition*, by Triebel and Singh

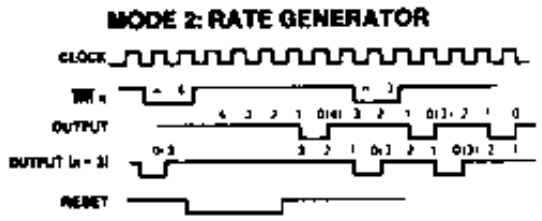
8253 modes

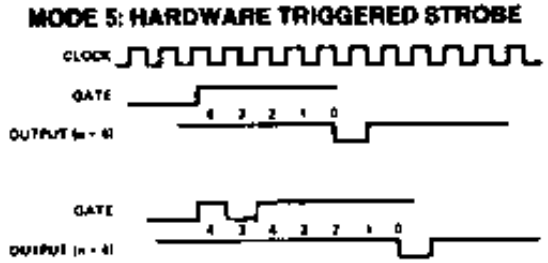












Taken from *The 8088 and 8086 Microprocessors, 2d Edition*, by Triebel and Singh

TimerDemo: set up and initiate count

```
; timerdemo          an 8088 assembly language program to illustrate
; use of timer 2 of the 8253 inside a PC

mov dx, pcctcw              ; set up dx for control word
mov al, 10110100xb          ; counter 2, least/most, mode2, binary
out dx,al

mov dx, pcct2               ; set up dx to load initial count
mov al, 0ffh                ; initial count will be 0ffffh
out dx, al                  ; low byte of count
out dx, al                  ; high byte of count

mov dx, pcchb                ; this block changes only PB0 and causes a positive edge
in al, dx                   ; read current status of port b
and al, 11111110xb          ; pb0 = 0, others unchanged
out dx, al
or al, 1                    ; pb0 = 1, others unchanged
out dx, al                  ; positive edge starts count
```

TimerDemo: latch and read counters

```
mov dx, pcctcw                  ; set up dx for control word
mov al, 10000100xb  ; to latch counter 2
out dx,al

mov dx, pcct2                   ; this reads from the latch
                                                ; the counter continues to decrement
in al, dx                       ; low byte
mov bl, al
in al, dx                       ; high byte
mov bh, al
```

TimerDemo: housekeeping and end

```
call keywt

mov dx, pcchb                 ; "good housekeeping" set PB0 to 0 to disable count
in al, dx                     ; read current status of port b
and al, 11111110xb                        ; pb0 = 0, others unchanged
out dx, al                    ; count disabled

call godos
```

Ways to crash your computer -- Part 3

- Configure a timer other than Counter 2
  - Counters 1 and 0 are used for the DOS clock and to control the dynamic RAM refresh rate
  - If you set the refresh rate too low, memory forgets
- Change all the bits in the 8255's Port B
  - Bit 0 of Port B controls gate 2
  - What do the other bits control? Who knows?
  - Use AND and OR with masks to change only the bit you're interested in

Homework for Thursday, Sept. 23

- Write your own “benchmark” program using Timer2 of the PC. That is, find the time it takes to execute 100 instances of an instruction. Use this program to compare execution speed of two different computers.
- Submit an electronic copy of your source file and your *.com file. Also submit an electronic text file that reports execution times for 100 instances--state which instruction you investigated, and the processor and speed of the machines (if known).

Flow chart for homework

